Bill of Materials

A "bill of materials", often abbreviated to BOM, is what it says: it is a list of materials, in this case it is the list of all components and parts in the kit.

You can download a PDF version of the x0xb0x BOM here: x0xb0x_bom.pdf

Expanded Bill of Materials

Thanks to control voltage for putting this together.

Description	Qty	Part Number	Distributor	Part Locations:	
ABS Case	1	Pactec PT-10	Mouser, PacTec	kind of obvious	
Set of main and I/O panels	1	Custom		attached to the case	
Set of main and I/O PCBs	1	Custom		inside the case	
9VAC 300-500mA wall wart	1	Generic	Jameco	plugged in	
15/32" x 1/4" x 4-40 standoff	3	Generic	McMaster	mounted to main PCB	
3/8" x 1/4" x 4-40 standoff	4	Generic	McMaster	mounted to main PCB	
4-40 lock washer	7	Generic	McMaster		
4-40 washer	4	Generic	McMaster		
4-40 x 1/4" hex button	7		McMaster		
4-40 x 1/4" phillips button	7		McMaster		
4-40 x 1/4" phillips cheesehead (i swear its called that)	4	Comes with case			
Rubber feet	4	Comes with case		bottom of case	
Switch cap (white)	8	401-1155	Digikey		
Switch cap (black)	15	401-1152	Digikey		
Potentiometer knob	8	450-4760	Mouser		
Switch knob	2	5164-1510	Mouser		
Heat sink (TO-220)	1	532-574502b00	Mouser	mount to IC20	
40-pin DIP socket	1	571-3902625	Mouser	IC3* (socket into PCB, IC into socket)	

Description	**Qty**	**Part Number**	**Distributor**	**Part Locations:**	
8-pin DIP socket	1	571-3902612	Mouser	IC2* (ditto above)	
1/4" stereo jack	3	568-nys216	Mouser	MIXOUT, MIXIN, HEADPHONE	
1/8" jack	2	cp-3536n	Digikey	CV, GATE	
2.1mm power jack	1	cp-202a	Digikey	POWER	
USB type-B jack	1	154-2442	Mouser	USB	
MIDI/DINSYNC jack	4	cp-2350	Digikey	DIN1-DIN4	
Toggle switch	1	633-b12ab	Mouser	S1	
Rotary encoder	1	ct3011	Digikey	S2	
16 position rotary switch	2	gh3074	Digikey	S3, S4	
Tact switch	23	eg1821	Digikey	S5-S27	
Solid ~20-gauge wire. (Supply this yourself)	1	Generic		used to anchor S3, S4	
jumper ribbon cable	**FixMe** 6 wire OR 9 wire and 6 wire 4 OR 2 x 9c and 1x 6c	WM09-06, WM06-06, WM03-06	Digikey	J1-J4	
0.22" LED standoff	40	593-STD240B	Mouser	LED1*-LED40* (mounted onto LEDs)	
Red 5mm LED	40	Generic		LED1-LED40	
1N4001	4	Generic		D40-D43 (PowerSupply)	
1N4148	29	Generic		D1-D10(Sequencer); D24(VCF); D25, D30, D31(VCO); D27, D35, D36(VCA); D26, D28, D29, D34, D37(Envelope); D32, D33(Mixer); D44-D47(PowerSupply); D48(IOBoard);	
10 ohm 5% resistor	2	Generic		R186, R164 (IOBoard)	
22 ohm 5% resistor	2	Generic		R20 (IOBoard); R150 (Envelope)	
27 ohm 5% resistor	2	Generic		R6, R7 (IOBoard)	
100 ohm 5% resistor	7	Generic		R1(PowerSupply); R16(IOBoard); R95(VCF); R130, R132(VCA); R136(Mixer); R152(Envelope)	
220 ohm 5% resistor	5	Generic		R13-R15, R18, R19 (IOBoard)	
470 ohm 5% resistor	1	Generic		R12 (IOBoard)	
1K 5% resistor	4	Generic		R2 (PowerSupply); R17(IOBoard); R137(Envelope); R161(Mixer)	
1.5K 5% resistor	1	Generic		R8 (IOBoard)	
1.8K 5% resistor	1	Generic		R179 (PowerSupply)	

Description	Qty	Part Number	Distributor	Part Locations:	
2.2K 5% resistor	19	Generic		R21-R24(IOBoard); R67, R68, R69, R70, R71, R98, R108, R124, R125, R126(VCF); R104, R107(VCO); R133, R162(VCA); R178(PowerSupply)	
4.7K 5% resistor	2	Generic		R9(IOBoard); R135(Mixer)	
6.8K 5% resistor	1	Generic		R3 (PowerSupply)	
10K 5% resistor	94	Generic		R10, R11(IOBoard); R34, R36, R105, R101(VCO); R47, R61, R64, R65, R94, R96, R97, R109, R112, R115, R116(VCF); R142, R143, R144, R145, R148, R149(Envelope); R201-R234, R237-R270 (Sequencer)	
22K 5% resistor	10	Generic		R45, R60(VCO); R110, R111(VCF); R117, R146, R151(Envelope); R120, R129, R134(VCA)	
33K 5% resistor	1	Generic		R159 (Mixer)	
47K 5% resistor	4	Generic		R46(VCF); R155(Mixer); R119, R127(VCA)	
68K 5% resistor	1	Generic		R138 (Envelope)	
100K 5% resistor	24	Generic		R25-R28(IOBoard); R35, R93, R92, R59, R102, R118(VCO); R66, R72, R73, R99, R113, R114(VCF); R139, R140, R141(Envelope); R122(VCA); R153, R154, R156, R165(Mixer)	
220K 5% resistor	8	Generic		R62, R63(VCF); R103(VCO); R121, R128, R131(VCA); R235, R236(Sequencer)	
1MEG 5% resistor	2	Generic		R91(VCO); R200(Sequencer)	
1.5MEG 5% resistor	1	Generic		R123 (VCA)	
2.4K 1% resistor (red yellow black brown brown)	1	Generic		R5 (PowerSupply)	
5.6K 1% resistor (green blue black brown brown)	1	Generic		R4 (PowerSupply)	
24K 1% resistor (red yellow black red brown)	1	Generic		R106 (VCO)	
200K 1% resistor	17	Generic		R74-R90 (VCO)	
2K (202) trim potentiometer	1	306uc202b	Digikey	TM6 (PowerSupply)	
5K (502) trim potentiometer	1	306uc502b	Digikey	TM5 (VCO)	
50K (503) trim potentiometer	1	306uc503b	Digikey	TM4 (VCO)	

Description	Qty	Part Number	Distributor	Part Locations:	
500K (504) trim potentiometer	1	306uc504b	Digikey	TM3 (VCF)	
50K D (log) potentiometer	3	p3l2503	Digikey	VR3, VR5(VCF); VR8(Mixer)	
50K B (linear) potentiometer	2	p3k1503	Digikey	VR2 (VCO); VR7(VCA)	
50K B (linear) dual potentiometer	1	317-2202-50k	Mouser	VR4 (VCF)	
1Meg A (log) dual potentiometer	1	317-2102-1M	Mouser	VR6 (Envelope)	
1K PTC Thermistor	2	/732278	Newark/Farnell	R100A, R100B (VCO)	
.001uF (2A102K) capacitor	3	140-pm2a102k	Mouser	C34(VCO); C100, C101(Sequencer)	
.0068uF (2A682K) capacitor	1	140-pf2a682k	Mouser	C47 (Mixer)	
.01uF (2A103K) capacitor	5	140-pm2a103k	Mouser	C33, C10(VCO); C21, C20(VCA); C46(Mixer)	
.018uF (2A183K) polyester capacitor	1	140-pf2a183k	Mouser	C18 (VCF)	
.033uF (2A333K) polyester capacitor	5	140-pm2a333k	Mouser	C19, C24, C26(VCF); C36(VCA); C201(IOBoard)	
.047uF (2A473K) capacitor	1	140-pm2a473k	Mouser	C54 (Envelope)	
.068uF (2A683K) capacitor	1	140-pf2a683k	Mouser	C45 (Mixer)	
.1uF (2A104K) capacitor	3	140-pm2a104k	Mouser	C25, C27(VCF); C41(VCA)	
.1uF (104) ceramic	16	581-sr205e104m	Mouser	C1, C2, C4, C6(PowerSupply); C202, C204(IOBoard); C103, C104, C106-C113 (Sequencer)	
.22uF (2A224K) polyester capacitor	1	140-pm2a224k	Mouser	C35 (VCO)	
1uF electrolytic capacitor	13	140-xrl50v1.0	Mouser	C11(VCO); C13, C14, C15, C17, C22, C23, C29(VCF); C38(VCA); C58, C59, C56(Mixer); C105(Sequencer)	
1uF (105) tantalum capacitor	2	399-1429	Digikey	C62(Envelope); C42(VCA)	
2.2uF electolytic capacitor	1	140-xrl50v2.2	Mouser	C51 (Mixer)	
10uF electrolytic capacitor	13	140-xrl16v10	Mouser	C60, C61(PowerSupply); C31, C32(VCO); C16, C30(VCF); C72(Envelope); C37, C40(VCA); C50, C52(Mixer); C203, C205 (IOBoard)	

Description	Qty	Part Number	Distributor	Part Locations:	
47uF electrolytic capacitor	4	140-xrl16v47	Mouser	C28(VCF); C55(Envelope); C44, C53(Mixer)	
100uF 10V electrolytic capacitor	3	140-xrl25v100	Mouser	C48, C49(Mixer); C102(Sequencer)	
100uF 25V electrolytic capacitor	2	140-xrl10v100	Mouser	C7, C8 (PowerSupply)	
1000uF electrolytic capacitor	1	140-xrl10v1000	Mouser	C43 (Mixer)	
2200uF electolytic capacitor	2	140-xrl16v2200	Mouser	C3, C5 (PowerSupply)	
4066 analog switch	1	Generic		IC12 (VCO)	
74AC174	1	Generic		IC9 (VCO)	
74HC126	1	Generic		IC1 (Sequencer) incorrect in fab as 74ac129	
74AC165	3	Generic		IC16-IC18 (Sequencer)	
74HC595	5	Generic		IC4-IC8 (Sequencer) incorrect in fab as 74ac595	
2SA733P, TO-92 PNP transistor	10	2SA733P	Mouser	Q8, Q27(VCO); Q9, Q10(VCF); Q36, Q38(Envelope); Q1*, Q2*, Q31(VCA); Q33(mixer)	
2SC536F, TO-92 NPN transistor	24	Rare		Q29, Q30, Q25, Q24(VCO); Q11, Q13, Q14, Q15, Q16, Q17, Q18, Q19, Q20, Q23(VCF); Q35, Q37, Q40, Q41(Envelope); Q3*, Q4*, Q32(VCA); Q34(Mixer); Q50(IOBoard); Q5(Sequencer)	
2SK30 JFET	2	Rare		Some kits have a 2SK30AY and a 2SK30AO, see the instructions for which one to put where.	
AN6562 8-DIP dual Op-Amp	2	Rare		IC23 (PowerSupply); IC11 (VCO)	
2SC2291 5-SIP matched pair with common base	1	Rare		Q22 (VCF)	
2SC1583 5-SIP matched pair with common emitter	3	Rare		Q12, Q21(VCF); Q26(VCO)	
LA4140 9-SIP	1	Rare		IC14 (Mixer)	
BA6110 9-SIP	1	Rare		IC15B* (VCA)	
BA662A 9-SIP	(optional)	Extremely rare		IC15A* (VCA) - highly unlikely	
ATmega162 microcontroller	1	Atmega162-40PI	Digikey (for programmed version, Adafruit)	IC3 (Sequencer)	

Description	Qty	Part Number	Distributor	Part Locations:	
25C33 EEPROM	1	cat25c33p	Mouser	IC2 (Sequencer)	
7805 voltage regulator (TO-220)	1	512-mc78m05ct	Mouser	IC20 (PowerSupply)	
7806 voltage regulator (TO-92)	1	513-Njm78l06a	Mouser	IC21 (PowerSupply)	
LM336Z-5.0 5V voltage reference (TO-92)	1	lm336z-5.0	Digikey	IC22 (PowerSupply)	
4N37 optoisolator	1	512-4n37	Mouser	IC24 (IOBoard)	
FT232 USB chip	1	FT232BM	Mouser	IC25 (IOBoard)	
6MHz ceramic resonator	1	815-awcr-6.00md	Mouser	XTL2 (IOBoard)	
16MHz ceramic resonator	1	815-awcr-16.00md	Mouser	XTAL1 (Sequencer)	